

MUHAMMED RAIHANUL HOQUE

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PROFILE

Master's student in Electrical Systems Engineering focusing on Analog/Mixed-Signal (AMS) IC design and verifications. Hands on experience in designing high speed analog front end from schematic to layout and physical verification (DRC/LVS/PEX) in Cadence Virtuoso. Trying to explore the advanced semiconductor industry through continuous learning.

EDUCATION

M.S. in Electrical Systems Engineering

Paderborn University, NRW, Germany

German Grade: 2.0

- **Master's Thesis:** Design of a Low-Noise Fully Differential Transimpedance Amplifier (TIA) with DC Offset Compensation for Coherent IQ Receiver (Grade: 1.3).
- **Key Results:** Achieved 80 GHz bandwidth (160 Gb/s NRZ) and 14 pA/ $\sqrt{\text{Hz}}$ input-referred noise, 37 dB dynamic range, and 4mA DC offset compensation using IHP 130 nm SiGe BiCMOS.
- **Workflow:** Topology definitions, schematic design, corner simulations, layout design (shielding, matching, symmetry, guard ring), DRC and LVS verifications, PEX extraction, EM simulation (RFPro).
- **Coursework:** Circuit and System Design, IC Design for Wireline Communications, RF IC Design, Fields and Waves, Advanced Systems Theory.

B.S. in Electrical and Electronic Engineering

Ahsanullah University of Science and Technology, Dhaka, Bangladesh

German Grade: 2.3

- **Coursework:** Electronics I & II, Solid State Devices, Electromagnetics, Digital Electronics, Circuit Theory, Control System.

PROJECTS

Two-Stage Miller Compensated OpAmp Design using g_m/I_D methodology (130nm BiCMOS):

- Key Specs: DC Gain: 52 dB, Phase Margin: 67°, Unity Gain Frequency (UGF): 60 MHz.
- Performance: Achieved PSRR of 60/70 dB (Pos/Neg) and CMRR of 60 dB with low power consumption of 550 μW .

RF/Optical Receiver Design (IHPs 130nm BiCMOS G3)

- Design of a low noise differential TIA.
- Comparison between feedforward and shunt feedback TIA in terms of noise and BW performance.

Dielectric Slab Waveguide Mode Solver

- Designed a quasianalytic mode solver for anisotropic thin film lithium niobate (TFLN) slabs.

TECHNICAL SKILLS

Circuit Knowledge: Differential pairs, current mirror, cascode amplifier, OpAmp, CTLE, Gilbert cell, VGA.

EDA Tools: Cadence Virtuoso (Layout XL, ADE, Spectre), ADS, RFPro, LTspice.

Physical Verification: Assura/PVS DRC, LVS, PEX extraction.

Scripting/OS: SKILL, Python, MATLAB, Linux environment.

LANGUAGES

Bengali: Native **English:** C1 **German:** Basic (Learning through academic medium)